

An Introduction to Multiobjective Optimization

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Single Objective Optimization

Inputs

- **Search space:** Set of all feasible solutions,

$$\mathcal{X}$$

- **Objective function:** Quality criterium

$$f : \mathcal{X} \rightarrow \mathbb{R}$$

Goal

Find the best solution according to the criterium

$$x^* = \operatorname{argmax} f$$

But, sometime, the set of all best solutions, good approximation of the best solution, good 'robust' solution...

Context

Black box Scenario

We have only $\{(x_0, f(x_0)), (x_1, f(x_1)), \dots\}$ given by an "oracle"
No information is either not available or needed on the definition of objective function

- Objective function given by a computation, or a simulation
- Objective function can be irregular, non differentiable, non continuous, etc.
- (Very) large search space for discrete case (combinatorial optimization), *i.e.* NP-complete problems
- Continuous problem, mixt optimization problem

Real-world applications

Typical applications

- Large combinatorial problems:
Scheduling problems, planing problems, DOE,
"mathematical" problems (Firing Squad Synchronization Pb.), etc.
- Calibration of models:
Physic world $\Rightarrow$ Model(params) $\Rightarrow$ Simulator(params)
 $\text{Model(Params)} = \text{argmin}_M \text{Error(Data, } M)$
- Shape optimization:
Design (shape, parameters of design)
using a model and a numerical simulator

Search algorithms

Principle

Enumeration of the search space

- A lot of ways to enumerate the search space
- Using random sampling: Monte Carlo technics
- Local search technics:



Search algorithms



- **Single solution-based:** Hill-climbing technics, Simulated-annealing, tabu search, Iterative Local Search, etc.
- **Population solution-based:** Genetic algorithm, Genetic programming, ant colony algorithm, etc.

Design components are well-known

- Probability to decrease,
- Memory of path, of sub-space
- Diversity of population, etc.

Research question: Parameters tuning

- One Evolutionary Algorithm key point:
Exploitation / Exploration tradeoff
- One main practical difficulty:
Choose operators, design components, value of parameters, representation of solutions
- Parameters setting (Lobo et al. 2007):
 - Off-line before the run: *parameter tuning*,
 - On-line during the run: *parameter control*.

One practical and theoretical question

How to combine correctly the design components according to the problem (in distributed environment...) ?

Research question: Expensive optimization

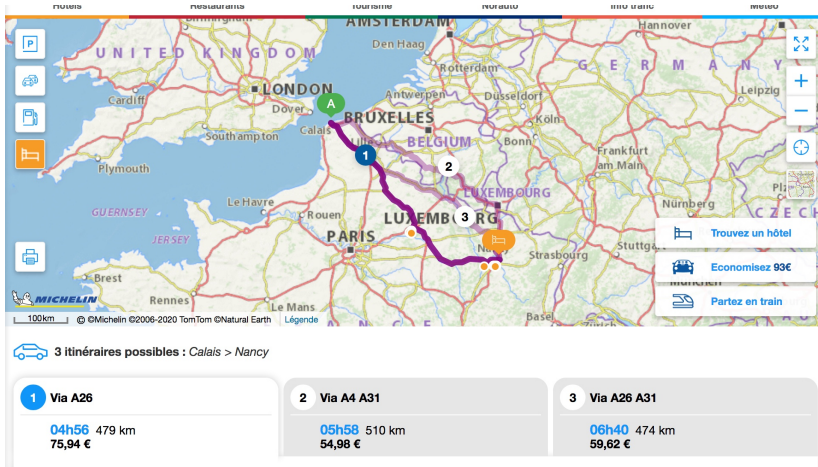
- Objective function based on a simulation:
Expensive computation time
- One main practical difficulty:
With few computation evaluation, choose operators, design components, value of parameters, ...
- Two main approaches:
 - Approximate objective function: *surrogate model*,
 - Parallel computation: *distributed computing*.

One practical and theoretical question

How to combine correctly the design components
with low computational budget
according to the problem in distributed environment... ?

Small example

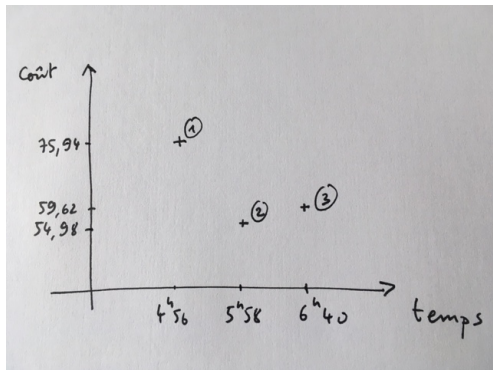
Road trip between Calais and Nancy



Which one is better ?

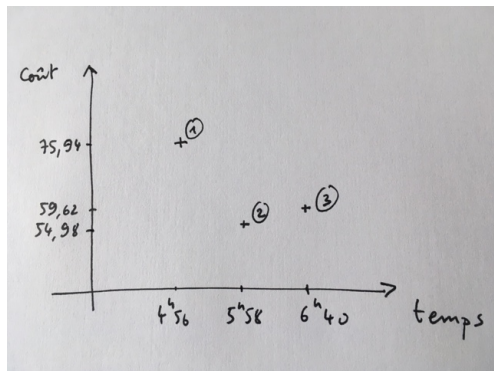
Small example

Road trip between Calais and Nancy



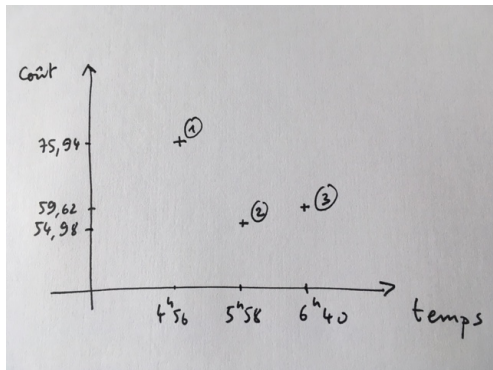
Small example

Road trip between Calais and Nancy



- According to time objective, 1 is better
- According to cost objective, 2 is better
- But, 2 is better than 3 for both objectives.

Pareto dominance



- 1 and 2 are incomparable
- 1 and 3 are incomparable
- 2 is better than 3

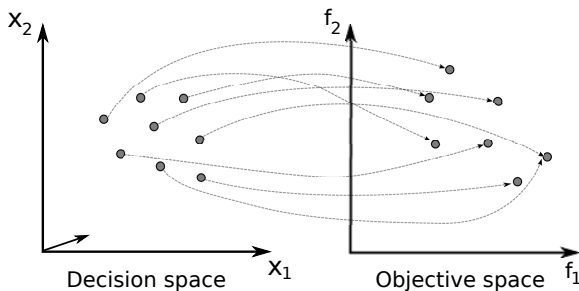
Pareto dominance

- 2 dominates 3
- 3 is dominated by 2

Multiobjective optimization

Multiobjective optimization problem

- $\mathcal{X}$: set of feasible solutions in the **decision space**
- $M \geq 2$ objective functions $f = (f_1, f_2, \dots, f_M)$ (to maximize)
- $\mathcal{Z} = f(\mathcal{X}) \subseteq \mathbb{R}^M$: set of feasible outcome vectors in the **objective space**

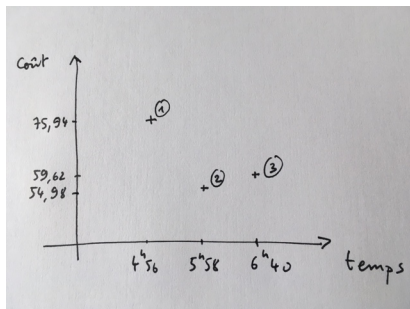


Pareto dominance definition

Pareto dominance relation (maximization)

A solution $x \in \mathcal{X}$ **dominates** a solution $x' \in \mathcal{X}$ ($x' \prec x$) iff

- $\forall i \in \{1, 2, \dots, M\}, f_i(x') \leq f_i(x)$
- $\exists j \in \{1, 2, \dots, M\}$ such that $f_j(x') < f_j(x)$

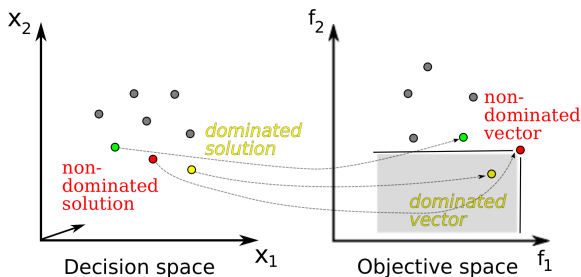


Pareto Optimale solution

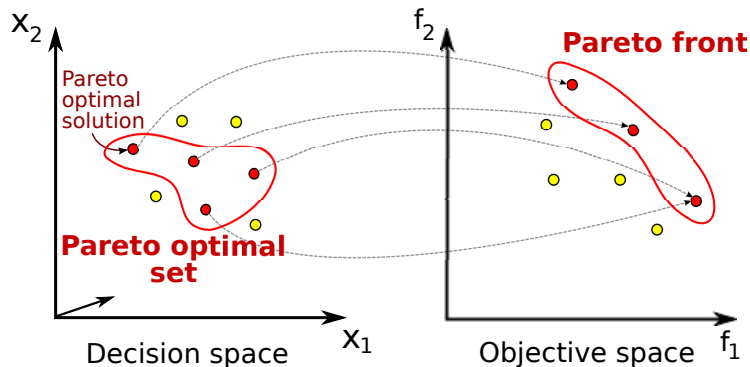
Definition: non-dominated solution

A solution $x \in \mathcal{X}$ is non-dominated (or Pareto optimal, efficient) iff

$$\forall x' \in \mathcal{X} \setminus \{x\}, x \not\leq x'$$



Pareto set, Pareto front



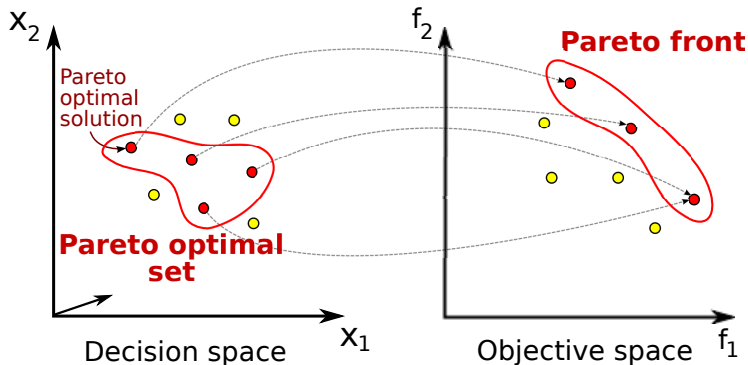
Vilfredo Pareto (1848 - 1923)

source: wikipedia

Multiojective optimization goal

Goal

Find the **Pareto Optimal Set**,
or a **good approximation** of the Pareto Optimal Set
And not a single solution for a single aggregated objective



How to compute non dominated solutions from a set?

Filter by dominance relation with a basic algorithm: see exercise 2

Input: solution_set, the set of solutions to filter by dominance

Output: non_dominated_solutions, the set of non-dominated solutions

```
non_dominated_solutions  $\leftarrow \emptyset$ 
```

```
for solution  $\in$  solution_set do
```

```
    s  $\leftarrow$  first solution of solution_set
```

```
    while s  $\neq$  NULL && solution is not dominated by s do
```

```
        s  $\leftarrow$  next solution of solution_set
```

```
    end while
```

```
    if s  $\neq$  NULL then
```

```
        non_dominated_solutions  $\leftarrow$  non_dominated_solutions  $\cup$  { solution }
```

```
    end if
```

```
end for
```

```
return non_dominated_solutions
```

Time complexity: $\mathcal{O}(m^2 \times d)$

where m is the size of solutions set, and d the dimension of objective space

Challenges

- **Search space:**
many variables, heterogeneous, dependent variables
- **Objective space:**
many, heterogeneous, expensive objective functions
- **NP-completeness:**
deciding if a solution is Pareto optimal is difficult
- **Intractability:**
number of Pareto optimal solutions grows exponentially with problem dimension

Why multiobjective optimization?

Position of multiobjective optimization: Decision making

- No **a priori** on the importance/weights of the different objectives
- **a posteriori** selection by a decision marker:
Selection of one Pareto optimal solution after a deep study of the possible solutions.

Methodology

Typical methodology with MO optimization

- 1 Define **decision variables**
- 2 Define all potential **objective**
- 3 Define **constraints** (soft/hard/objective)
- 4 Choose/design a relevant multiobjective **algorithm**
- 5 Search for an approximation of **Pareto optimal** solutions set
- 6 **Analyse/visualize** the solutions set

Loop between 1 to 6...

Multi-objective, many-objective optimization

Approximative definition

- Multi-objective: 2, 3 or 4 objectives
- Many-objective: 4, 5 and more objectives

Number of Pareto optimal solutions

Suppose that:

- Probability to improve: p (for all objective),
- Objective are independent.

Probability to be non-dominated for M objectives is:

Multi-objective, many-objective optimization

Approximative definition

- Multi-objective: 2, 3 or 4 objectives
- Many-objective: 4, 5 and more objectives

Number of Pareto optimal solutions

Suppose that:

- Probability to improve: p (for all objective),
- Objective are independent.

Probability to be non-dominated for M objectives is: $1 - (1 - p)^M$

Multi-objective, many-objective optimization

Approximative definition

- Multi-objective: 2, 3 or 4 objectives
- Many-objective: 4, 5 and more objectives

Number of Pareto optimal solutions

Suppose that:

- Probability to improve: p (for all objective),
- Objective are independent.

Probability to be non-dominated for M objectives is: $1 - (1 - p)^M$

Intuitive goals

Convergence toward the front, and diversity of the solutions.

Many-objective: convergence "easy", diversity "hard"

Note: objective correlation is also important.

Performance assessment

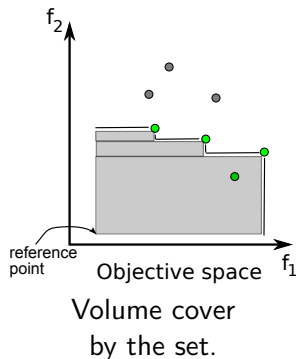
Quality of the approximation of the Pareto front [1]

- Goal: indicator function related to the quality of the set.
- No universal indicator

Indicator functions

- Hypervolume indicator
- Epsilon indicator
- Inverted Generational Distance (IGD)
- Attainment function

Hypervolume (I_H)



Properties

- Compliant with the (weak) Pareto dominance relation
 $\rightarrow A \prec B \Rightarrow I_H(A) \leq I_H(B)$
- A single parameter:
the reference point
- Minimal solution-set maximizing I_H
 $\rightarrow$ subset of the Pareto optimal set

$$\arg \max_{\sigma \in \Sigma} I_H(\sigma)$$

Slide to draw

(additive) Epsilon indicator (I_ϵ)

Definition

Smallest coefficient ϵ to translate the set A to "cover" each point of the set B

For maximization, minimal ϵ value such that:

$$\forall z_b \in B, \exists z_a \in A \text{ such that } z_b \prec z_a + \epsilon$$

Properties

- Compliant with the (weak) Pareto dominance relation (using Pareto front)
 $\rightarrow A \prec B \Rightarrow I_\epsilon(A) \leq I_\epsilon(B)$
- A parameter: the reference set

Slide to draw

Inverted Generational Distance (IGD)

Definition: Generational Distance (GD) of set A

Average over all solutions $a \in A$ of the distance between solution a and the closest solution in a reference set R :

$$IG(A, R) = \frac{1}{|A|} \sum_{a \in A} \min_{r \in R} \text{dist}(a, r)$$

where $\text{dist}(a, r)$ is the euclidian distance in objective space between solution a , and r .

Definition: Inverted Generational Distance (IGD)

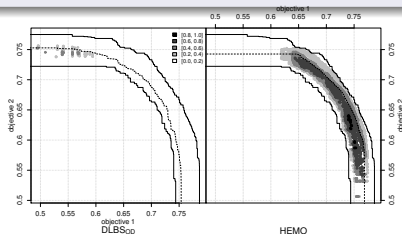
$$IGD(A, R) = IG(R, A)$$

Slide to draw

Attainment function

Definition

Probability to reach a point in objective space



Tools

Empirical Attainment Function (EAF) Tools, Manuel López-Ibáñez:
<https://mlopez-ibanez.github.io/eaf/>

Manuel López-Ibáñez, Luís Paquete, and Thomas Stützle. Exploratory Analysis of Stochastic Local Search Algorithms in Biobjective Optimization. In Experimental Methods for the Analysis of Optimization Algorithms, 2010.



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Eckart Zitzler.

A tutorial on the performance assessment of stochastic
multiobjective optimizers.

In *Third International Conference on Evolutionary
Multi-Criterion Optimization (EMO 2005)*, volume 216, page
240, 2005.